$$= -\frac{1}{t^2}\cdot\frac{1}{2at}$$

$$= -\frac{1}{2at^3}$$

(iii) $y = x \sin x$

$$\frac{dy}{dx} = x\cos x + \sin x$$

$$\frac{d^2y}{dx^2} = -x\sin x + \cos x + \cos x$$

$$= 2\cos x - x\sin x$$

Example 5.47

If $y = A\sin x + B\cos x$, then prove that $y_2 + y = 0$

Solution

$$y = A\sin x + B\cos x$$

$$y_1 = A\cos x - B\sin x$$

$$y_2 = -A\sin x - B\cos x$$

$$y_2 = -y$$

$$y_2 + y = 0$$

Exercise 5.9

1. Find y_2 for the following functions

   (i) $y = e^{3x+2}$

   (ii) $y = \log x + a^x$

   (iii) $x = a\cos\theta,\ y = a\sin\theta$

   IBFYV9

2. If $y = 500e^{7x} + 600e^{-7x}$, then show that $y_2 - 49y = 0$

3. If $y = 2 + \log x$, then show that $xy_2 + y_1 = 0$

4. If $y = a\cos mx + b\sin mx$, then show that $y_2 + m^2y = 0$

5. If $y = (x + \sqrt{1+x^2})^m$, then show that $(1+x^2)y_2 + xy_1 - m^2y = 0$

6. If $y = \sin(\log x)$, then show that $x^2y_2 + xy_1 + y = 0$.

Exercise 5.10

Choose the correct answer

1. If $f(x) = x^2 - x + 1$, then $f(x+1)$ is

   (a) x^2 (b) x

   (c) 1 (d) $x^2 + x + 1$

2. If $f(x) = \begin{cases} x^2 - 4x & \text{if } x \geq 2 \\ x + 2 & \text{if } x < 2 \end{cases}$, then $f(5)$ is

   (a) -1 (b) 2

   (c) 5 (d) 7

3. If $f(x) = \begin{cases} x^2 - 4x & \text{if } x \geq 2 \\ x + 2 & \text{if } x < 2 \end{cases}$, then $f(0)$ is

   (a) 2 (b) 5

   (c) -1 (d) 0

4. If $f(x) = \frac{1-x}{1+x}$, $x > 1$, then $f(-x)$ is equal to

   (a) $-f(x)$ (b) $\frac{1}{f(x)}$

   (c) $-\frac{1}{f(x)}$ (d) $f(x)$

5. The graph of the line $y = 3$ is

   (a) Parallel to x-axis

   (b) Parallel to y-axis

   (c) Passing through the origin

   (d) Perpendicular to x-axis

6. The graph of $y = 2x^2$ is passing through

   (a) (0,0) (b) (2,1)

   (c) (2,0) (d) (0,2)

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